

SURYAGRID AI

Main Plan

Execution Roadmap

Four-Level Timeline and Detailed Workflow

Solar data and APIs → forecasting and ML → per-station generation → demand gap → deterministic reward/penalty policy → safe RL research → controlled integration

Author: Lekhan HR / SuryaGrid Project

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Repository baseline: commit 5bed3cd

Scope: complete handwritten main plan, not only substation DSM

Planning horizon: 22 weeks from approved kickoff

Evidence sources: both handwritten plan images, current repository, test/build results, model cards, README, and report evidence matrix. Unclear handwriting and unapproved monetary notes are identified as assumptions, never converted into facts.

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1 Executive Direction

| What this plan corrects

The project is not only the recently implemented substation/DSM workflow. The governing handwritten main plan contains **four levels**: data/API, forecasting and intelligence, policy/economics, and operational workflow. Current Phase 1 work is a strong foundation, but live settlement, production RL, and plant control remain gated.

L4 - Operational Workflow Six logical agents, APIs/tools, QA, dashboard, alerts, audit, and later plant integration

L3 - Policy and Economics Generation/demand gap, DSM, reward, penalty, discount, and safely gated RL

L2 - Forecasting and Intelligence Weather/cloud interpretation, statistical/ML forecasting, energy calculation, uncertainty

L1 - Data and APIs Solar/weather providers, station configuration, schedule, load, and measured telemetry

Figure 1: The four-level main plan reconstructed from the handwritten source.

| Critical path

Official rules and measured telemetry → reliable data foundation → site-level forecast validation → deterministic DSM/policy → secure operator workflow → shadow pilot.

| Safety boundary

Until official economics and measured outcomes exist, SuryaGrid is decision support. No unsupported rupee settlement, no autonomous tariffs, and no plant dispatch. RL starts offline and is advisory-only in shadow mode.

1.1 Reference delivery horizon

For a cross-functional team of 3–5 engineers with part-time product, grid-domain, and compliance support:

- **Minimum credible decision-support pilot:** approximately 14–18 weeks.
- **RL-assisted, gated shadow pilot:** approximately 20–24 weeks.
- **External schedule risk:** official tariff, data, and pilot-site agreements can add 4–8 or more weeks.

2 Source Interpretation and Scope Alignment

2.1 Governing sources

1. Main handwritten plan: C:\Users\lekhanhr\Downloads\IMG_20260611_162613.jpg.
2. Handwritten Phase 1 note: C:\Users\lekhanhr\Downloads\IMG-20260618-WA0001.jpg.
3. Verified repository state: README.md, docs/report/report_evidence_matrix.md, code, tests, and model cards.
4. Older PROJECT_PLAN.md and SURYAGRID_PHASE1_ADVANCED_IMPLEMENTATION.md are supporting interpretations only.

2.2 Main-plan intent

Table 1: Main-plan level interpretation.

Level	Handwritten intent	Execution interpretation
L1	API / Solcast and data inputs	Provider-neutral solar, weather, station, schedule, load, and telemetry foundation.
L2	Cloud/weather, S/M, ML, station generation	Data quality, solar/cloud/load forecasts, pvlib energy calculation, and uncertainty.
L3	Cost, incentive, penalty, reward, discount, RL	Approved deterministic policy first. RL only after real economics and measured telemetry.
L4	Workflow	Six logical agents, APIs/tools, QA, dashboard, alerts, audit, and later hardware/SCADA.

2.3 Phase reconciliation

The handwritten Phase 1 note asks for an irradiance nowcast, solar generation prediction, agreement/scheduled-MW comparison, DSM threshold classification, penalty/no-penalty risk, and a time-based generation/DSM timeline. The original Phase 1 specification implemented this first as a toy-data/formula MVP. In the main plan’s numbering, that work is closer to a **simulation/foundation subset**, not the complete live pilot.

The repository now extends that baseline with real weather, pvlib, trained models, source provenance, 344 substations, REST APIs, a Next.js interface, and deployment assets. For execution:

- **Completed foundation:** historical toy/formula MVP plus current real-data prototype.
- **Current state:** decision-support prototype, not settlement of record or plant control.
- **Next release:** controlled live/shadow pilot for 1–3 sites.
- **Later release:** validated reward/discount policy, conditional RL, and read-only plant integration before any control consideration.

2.4 Important interpretation cautions

- The main image explicitly marks L1, L2, L3, and L4. It is not a three-level-only plan.
- The unclear cloud role is named **Cloud/Weather Risk Agent**. The image does not reliably support the earlier “Cloudinary Agent” wording.

- S/M and one station-system acronym remain unresolved.
- Handwritten “25” and “15% discount” notes are exploratory, not approved business rules.
- Solcast is an intended source, but its current integration is not verified.

3 Verified Current Baseline

The following checkpoint was verified on 9 July 2026. Backend tests and lint were run; the frontend production build was run. No capability is upgraded based only on intended design.

Table 2: Current implementation checkpoint.

Capability	Status	Evidence and limitation
FastAPI backend and REST APIs	IMPLEMENTED	61 /api/v1 routes.
Automated verification	IMPLEMENTED	118 backend tests passed; Ruff passed.
Next.js operator interface	IMPLEMENTED	Production build passed; 14 application routes.
Live weather	PARTIAL	Open-Meteo operates; Solcast/NASA live failover is not wired.
Solar generation forecast	PARTIAL	pvlib plus ML; output is estimated, not measured actual.
Cloud-risk forecast	IMPLEMENTED	Model cards and inference path exist.
Substation context	PARTIAL	344 OSM substations; capacity absent; voltage incomplete.
DSM classification	PARTIAL	Decision-support framework; official monetary tariffs pending.
Reward/discount settlement	BLOCKED	Existing default amounts are illustrative and unapproved.
RL	BLOCKED	Research environment exists; production model card reports insufficient real data.
Authentication and tenancy	GAP	Plans/settings exist; enforcement is incomplete.
SCADA/plant telemetry	GAP	No verified connector or measured local PV/load.
Deployment	PARTIAL	Docker/single-instance path exists; managed AWS is unverified code.
Production operations	GAP	HTTPS, SLOs, monitoring, backup/restore need verification.

3.1 Current truth boundaries

- PV generation is estimated from irradiance; it is not metered Bengaluru site truth.
- Substation capacity is missing for all 344 OSM records; dependent calculations must block.
- India-wide load and PV models remain non-production for local Bengaluru decisions.
- No real rupee DSM result is permitted until an approved, effective-dated rule source is parsed, independently reviewed, and tested.
- The RL policy is not production-ready and must remain blocked.
- Terraform proves intended architecture, not an actually deployed managed AWS stack.

4 Target Operating Workflow

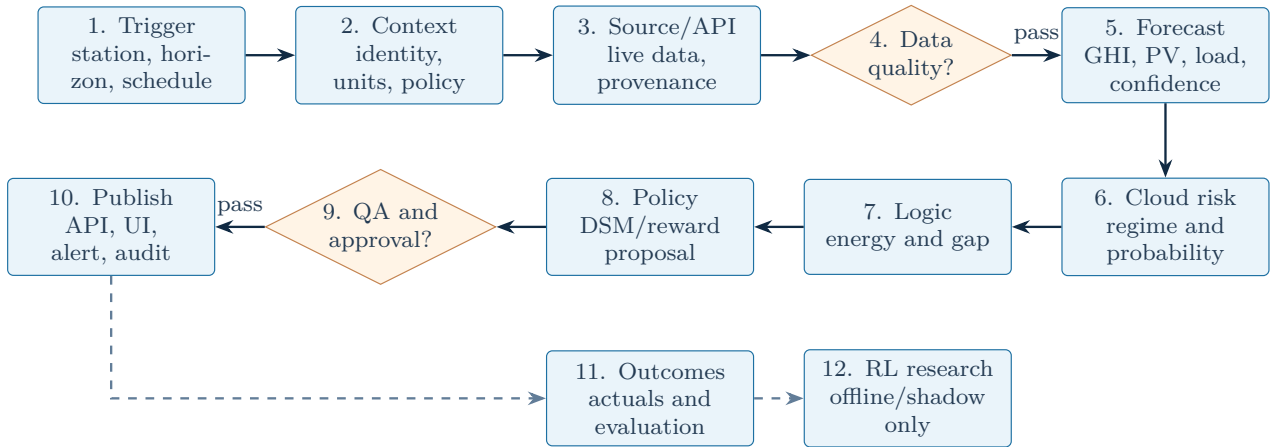


Figure 2: Target end-to-end agent workflow. The dashed path records actual outcomes for offline evaluation. Refetch and manual-review routes are specified in the detailed workflow.

4.1 Detailed workflow

A. Trigger and context

1. A scheduler or operator selects a station, horizon, schedule/request, and operating mode.
2. The Orchestrator creates a trace ID and resolves station metadata, timezone, units, nameplate, permissions, and effective policy version.
3. Missing required inputs are labelled and block dependent calculations.

B. Data acquisition and quality

4. The Source/API Agent retrieves live solar/weather data and recent measured actuals.
5. Provider records are normalized to a canonical interval schema.
6. Quality gates check freshness, completeness, range, units, timestamps, location, duplicates, source, and fallback state.
7. Raw payloads and normalized records are stored with immutable lineage.

C. Forecast and deterministic calculation

8. The Forecast Agent produces irradiance, PV generation, and, when local data permits, load forecasts with confidence intervals.
9. The Cloud/Weather Risk Agent produces cloud-regime and drop-risk probabilities.
10. The Logic/Calculation Agent uses versioned pvlib and energy formulas to calculate interval energy, schedule-versus-expected gap, and constraints.
11. When measured actual generation exists, forecast error and schedule deviation are calculated separately. A prediction is never labelled as measured actual.

D. Policy, QA, and publication

12. The Reward/Policy Agent applies the approved, effective-dated deterministic policy. Before approval, only risk/band output is allowed; rupee values remain disabled.
13. Any policy or RL recommendation is bounded, explainable, advisory, and human-approved.
14. QA checks source quality, model/formula version, bounds, confidence, policy version, and the complete calculation trace.
15. Failed checks route to refetch, recompute, manual review, or a blocked state.
16. Accepted results are persisted and exposed to the API, dashboard, timeline, alerts, reports, and audit log.
17. Actual outcomes return later for monitoring and, only after the research gate, offline RL.

5 Six Logical Agent Contracts

The repository can keep specialized modules, but product-level ownership should remain six clear responsibilities that map to the main plan.

Table 3: Logical agent contracts.

Agent	Input	Output	Failure behavior	Initial rule	release
Orchestrator	station, horizon, schedule, mode	state, trace, final result	bounded retry; honest block	no hidden agent decision	
Source/API	provider config, coordinates, interval	normalized series and provenance	failover or unavailable	two live providers before pilot	
Forecast	quality-approved features	GHI/PV/load forecast and interval	labelled formula fallback	site validation required	
Cloud/Weather Risk	weather and irradiance features	cloud risk and confidence	unknown-risk state	no invented camera input	
Logic/Calculation	forecast, schedule, constraints	energy and gap calculations	block missing dependencies	deterministic and versioned	
Reward/Policy	gap, policy, approved economics	band or bounded proposal	no-money blocked state	RL advisory only	

6 Execution Workstreams

6.1 A. Scope, policy, and evidence lock

When: Weeks 0–1 **Owner:** Product lead + energy-domain lead + technical lead **Depends on:** none

Tasks

- Approve the four-level scope and name each unclear handwritten concept.
- Select the first 1–3 pilot sites, personas, schedule source, settlement interval, and decision boundary.
- Disable or isolate unsupported monetary outputs.
- Freeze provenance labels, units, timezone rules, and missing-data behavior.
- Baseline APIs, model cards, tests, security gaps, and actual deployment topology.

Exit gate: signed scope/assumption register; no unapproved amount appears as regulatory truth; pilot success metrics and blocked states are agreed.

6.2 B. Official data and regulatory access

When: Weeks 2–5; external lead time may add 4–8+ weeks **Owner:** Product/domain lead + data engineer **Depends on:** A

Tasks

- Obtain applicable KERC/CERC/BESCOM DSM/tariff orders and legal review.
- Secure KPTCL/BESCOM/SLDC or pilot-site access to capacity, voltage, schedule, load, and interval actuals.
- Obtain plant nameplate, tilt/azimuth, inverter limits, meter definitions, interval, and timezone.
- Define licensing, consent, retention, sharing, and incident contacts.
- Write effective-dated source contracts and regulatory test cases.

Exit gate: approved regulatory source or explicit decision-support/no-money scope; at least one pilot site supplies measured PV and schedule data; data ownership and quality targets are recorded.

6.3 C. L1 production data foundation

When: Weeks 3–7 **Owner:** Backend/data team **Depends on:** A; overlaps B

Tasks

- Add a second live provider; integrate Solcast if contractually required.
- Add provider health, persistent quotas, bounded retries, circuit breaking, and failover.
- Add immutable raw archive and idempotent normalized ingestion.
- Add late-data, duplicate, gap, range, timezone, unit, and quarantine controls.
- Operate backfill and live ingestion as observable jobs.
- Add time-series indexing/retention and source-to-feature lineage.
- Define station, schedule, actual-generation, load, and policy-version contracts.

Exit gate: 30 consecutive days of feed operation or approved accelerated window; completeness at least 98%; freshness target met; zero silent synthetic fallback; all outputs resolve to source, timestamp, location, unit, and transformation.

6.4 D. L2 forecast and calculation validation

When: Weeks 6–11 **Owner:** ML/energy team **Depends on:** B for truth; C for ingestion

Tasks

- Backtest irradiance and PV by site and season with time-based splits.
- Calibrate uncertainty and approve MAE/RMSE/bias/coverage thresholds.
- Validate cloud-risk output and its effect on confidence.
- Build local load forecasting only after local truth exists.
- Keep predicted generation, measured actual, schedule, and demand separate.

- Version pvlib assumptions, loss factors, inverter limits, and formulas.
- Add drift, missing-feature, bounds, champion/challenger, and rollback tests.

Exit gate: signed site-level model validation; uncertainty visible in API/UI; labelled deterministic fallback or honest block, never fabricated actuals.

6.5 E. L3 deterministic DSM and incentive policy

When: Weeks 8–12 **Owner:** Domain lead + backend engineer + compliance reviewer **Depends on:** B and D

Tasks

- Encode effective-dated rule profiles with regulator, region, denominator, interval, slabs, and source citation.
- Test boundaries, over/under generation, aggregation, and unit conversion.
- Separate forecasted DSM risk from settlement based on measured actuals.
- Define rewards and consumer discounts only after legal/business approval.
- Produce immutable calculation trace and manual recomputation worksheet.
- Require dual approval for rule publication and immediate rollback.

Exit gate: golden regulatory suite passes; independent recomputation matches; unapproved profiles emit risk/band only and never currency.

6.6 F. L4 secure operator workflow and integration

When: Weeks 10–14 **Owner:** Frontend/backend/platform team **Depends on:** C; final acceptance uses D/E

Tasks

- Implement JWT/OIDC, RBAC, tenant/site isolation, audit, and enforced rate limits.
- Make one station selection drive weather, solar, cloud, generation, DSM, source, and trace views.
- Show measured versus predicted state, missing data, confidence, and blocked reasons.
- Add alerts/reports with acknowledgment and escalation.
- Implement one read-only pilot connector using the site's approved interface.
- Keep all plant control commands disabled during the first pilot.

Exit gate: UAT passes normal, fallback, stale, missing, and blocked scenarios; security tests prove isolation; connector replays data without controlling equipment.

6.7 G. Safe RL research gate

When: Weeks 12–17, conditional **Owner:** ML/research lead + domain reviewer **Depends on:** measured PV/load, approved economics, deterministic baseline

Tasks

- Define state, action, constraints, reward, simulator, and forbidden actions.
- Train only on approved historical/simulated data with leakage checks.

- Compare against no-action and deterministic baselines.
- Stress-test rare weather, missing data, economic extremes, and policy changes.
- Expose recommendations only in shadow mode with bounds, reason, confidence, and rollback.
- Record proposal, human decision, outcome, and model/policy version.

Exit gate: offline policy improves agreed metrics with zero constraint violations; domain/risk sign-off; RL remains advisory and cannot set tariffs, money, or dispatch.

If dependencies are absent at Week 12, pause this workstream without delaying the decision-support pilot.

6.8 H. Platform hardening and controlled pilot

When: Hardening Weeks 14–18; pilot Weeks 19–22 **Owner:** Platform/SRE + QA + pilot owner
Depends on: C–F; G only for RL-assisted pilot

Tasks

- Verify HTTPS/domain, secrets, dependency scanning, CI/CD, and environment separation.
- Add logs, metrics, traces, data-quality alerts, drift alerts, and SLO dashboards.
- Test backup/restore, failover, provider outage, rollback, and incident response.
- Define and test availability, latency, and data-freshness targets.
- Run 2–4 weeks of shadow operation and compare forecasts with measured actuals.
- Hold production-readiness review and record accepted residual risk.

Exit gate: no critical security issue; restore and rollback drills pass; pilot metrics and operator sign-off pass; go/extend/stop decision recorded.

7 Integrated Timeline and Gates

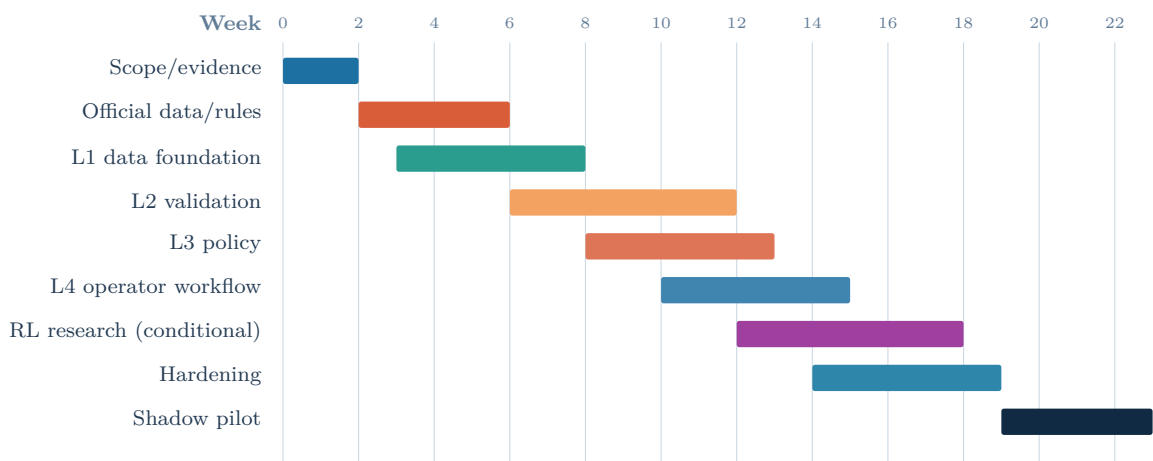


Figure 3: Dependency-based reference schedule. Week 0 is approved kickoff.

Table 4: Integrated phase gates.

Week	Main work	Parallel work	Gate
0–1	Scope and evidence lock	Quarantine unsupported money	G0 Scope
2–5	Official rules and telemetry access	Provider/data contract design	G1 Source
3–7	L1 ingestion, quality, lineage	Security foundation	G2 Data
6–11	L2 site forecast validation	UI trace and provenance	G3 Model
8–12	L3 deterministic policy	Operator workflow	G4 Policy
10–14	L4 auth, UI, read-only connector	Alerts and UAT	G5 Pilot-ready
12–17	Conditional RL research	Hardening	G6 RL shadow
14–18	SRE/security hardening	Incident and restore drills	G7 Operations
19–22	Controlled 1–3-site shadow pilot	Drift and operator review	G8
			Go/extend/stop

7.1 Dependency rules

- Frontend polish cannot compensate for missing source truth.
- Reward, discount, and currency cannot precede approved policy.
- RL cannot precede measured outcomes and a deterministic baseline.
- Control integration cannot precede read-only validation, security, audit, and human approval.
- A blocked dependency produces a smaller honest release; it does not permit fabricated data.

8 Acceptance Scorecard

Table 5: Pilot acceptance criteria.

Area	Required acceptance evidence
Data	At least 98% completeness in the agreed window; freshness SLO met; 100% of outputs carry provenance.
Forecast	Site/season metrics meet signed thresholds; uncertainty is calibrated and visible.
Calculation	Golden energy/DSM cases are reproducible; units, formula, and policy version explicit.
Safety	No unsupported rupee output; no autonomous dispatch; missing inputs block honestly.
Security	Auth, RBAC, site isolation, rate limits, secrets, audit, and HTTPS are verified.
Reliability	Provider outage, retry, backup/restore, rollback, and stale-data tests pass.
Operator UX	User completes station-to-decision flow and understands confidence and limitations.
RL (optional)	Offline improvement over baseline with zero constraint violation; advisory-only shadow mode.

9 Risks and Responses

Table 6: Execution risks and planned responses.

Risk	Effect	Response
Official rules delayed	No monetary settlement	Keep risk bands only; release decision support.
Measured PV/load unavailable	Forecast and RL cannot be locally validated	Make pilot data agreement the top commercial task.
Solcast unavailable/costly	Source and quota risk	Provider-neutral contract and second live fall-back.
OSM station fields incomplete	Loading/voltage calculations blocked	Obtain official capacity/voltage and preserve nulls.
India-model domain shift	Misleading local output	Keep non-production and retrain on pilot-site truth.
Hard-coded money reaches UI	Regulatory and credibility harm	Quarantine, label, and add tests forbidding unsupported currency.
RL optimizes unsafe proxy	Financial/operational harm	Constrained actions, baselines, stress tests, shadow mode, and human approval.
Security stays aspirational	Cross-site data exposure	Make enforcement and isolation tests a pilot gate.
Deployment differs from IaC claims	Reliability gap	Verify actual topology and label code-only assets.
Ambiguous handwriting	Wrong product decomposition	Maintain an assumption register and obtain owner confirmation.

10 Deliverables

1. Approved scope and assumption register.
2. Data-source, station, schedule, actual, load, and policy contracts.
3. Reliable multi-provider ingestion with raw archive, quality gates, and lineage.
4. Site-validated solar/load forecasts with model cards and uncertainty.
5. Versioned deterministic energy and DSM library with golden tests.
6. Secure operator dashboard, alerting, audit, and read-only connector.
7. Operations package: CI/CD, observability, SLOs, backup/restore, rollback, incident runbook.
8. Shadow-pilot report with data quality, accuracy, operator findings, and go/stop decision.
9. Conditional RL evaluation report and advisory shadow policy.

11 Immediate Next 10 Working Days

1. Approve this roadmap and the permanent context prompt.
2. Confirm the unclear cloud role, S/M term, regulator, settlement interval, and pilot boundary.
3. Select 1–3 pilot sites and request nameplate, schedule, meter, PV actual, and load samples.
4. Disable or isolate unsupported rupee outputs.
5. Make one frontend station selection drive the full agent result and source trace.
6. Verify hosted topology, HTTPS, production database content, and backup state.
7. Draft provider/telemetry contracts and the official-rule evidence checklist.
8. Open the regulatory/data-access workstream because it is the longest external dependency.

12 Permanent Context and Open Decisions

The durable prompt is stored at: D:\SuryagridAI\docs\SURYAGRID_MASTER_PLAN_CONTEXT_PROMPT.md. It records the main image interpretation, Phase 1 reconciliation, verified current baseline, non-negotiable truth/safety rules, execution order, and future-agent response requirements.

Owner decisions still required:

- Is Solcast mandatory, or is a provider-neutral source layer acceptable?
- Does the cloud role mean weather risk, cloud imagery/camera, or a media service?
- What do S/M and the unclear station-system acronym mean?
- Which Karnataka market participant and DSM regulation apply?
- What are the settlement interval, schedule source, and under/over-generation treatment?
- Which 1–3 sites can provide measured PV, schedule, load, and plant configuration?
- Is the first live release decision support only, or is any control action expected?

Final accuracy note

The two handwritten images are the primary intent sources, but parts of the main image are illegible. “Cloud/Weather Agent” is a functional label, not a claim that the note says “Cloudinary.” The values 25 and 15% are not approved requirements. All durations are planning estimates and must be re-baselined after pilot data and regulatory access are confirmed.
